

Course Outline

1. Course Identity

A. Course as listed in CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MDS5117
Course title (English)	Data Visualization
Course title (Chinese)	数据可视化
Units	3
Language of Instruction	English
Description (English)	Data visualization is the process used to communicate information clearly in graphical form. Topics covered include overview of the concepts and models used to visualize data, data models, graphical perception, and techniques for visual encoding and interaction, data visualization tools, data visualization projects using business-focused data from sources such as social media, ecommerce, marketing, etc.
Description (Chinese)	数据可视化是用于以图形形式清楚地传达信息的过程。涵盖的主题包括用于可视化数据、数据模型、图形感知和可视化技术的概念和模型的概述编码和交互、数据可视化工具、使用来自源的以业务为中心的数据的数据可视化项目

B. Corresponding course in CUHK

Please give details of the *closest* corresponding course in CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course in CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

2. Prerequisites / Co-requisites

Please state prerequisites and co-requisites, in terms of courses in CUHK-Shenzhen* or any other requirements (e.g., having taken certain subjects in high school).

(* Because course codes may not yet be stable, please provide both course code and course title.)

A. Prerequisites

No

B. Co-requisites

No

3. Learning Outcomes

- **Understand Core Concepts:** Gain a deep understanding of visualization techniques and visual analytics, including their theoretical foundations, purpose, and role in supporting data-driven decision-making.
- **Identify and Evaluate Key Issues:** Learn to critically assess the challenges and limitations in applying visualization techniques, including ethical considerations, data complexity, and audience-specific requirements.
- **Demonstrate Effective Tool Usage:** Develop hands-on expertise in industry-standard visualization tools such as Tableau, Power BI, and Python libraries, enabling the creation of impactful and actionable data visualizations.
- **Extract and Present Insights:** Learn how to extract meaningful insights from complex datasets and communicate these effectively through clear and compelling visual narratives tailored to diverse business stakeholders.
- **Develop Problem-Solving Skills:** Strengthen problem-solving abilities by leveraging data visualization and visual analytics to explore, analyze, and address complex business problems.
- **Bridge Theory and Practice:** Apply visualization techniques in real-world scenarios to connect theoretical concepts with practical business applications, ensuring students are prepared for the dynamic demands of the industry.

4. Course syllabus

The course will cover the following key topics related to Strategic Data Visualization for Business:

- o Introduction to Data Visualization and Design Principles
- o Data Preparation and Visualization Tools
- o Designing Dashboards and Interactive Visualizations

- o Storytelling with Data and Ethical Considerations
- o Project: Applying Visualization to Business Problems

By covering these topics, students will gain a solid understanding of data visualization principles and tools, along with practical skills to create impactful visualizations and dashboards. Through hands-on projects, they will learn to analyse real-world datasets, communicate insights effectively, and address business challenges using data-driven storytelling.

5. Assessment Scheme

Component/ method	% weight
Class participation	10
Individual Assignments	20
Mid-term exam*	15
Group Project	25
Final exam*	30

*Both exams are close-book

6. Grade descriptor

General

Grade	Description
A	The student demonstrates a good understanding of the subject matters and can demonstrate excellent performance in the individual assignments, group project and final examination
A-	The student demonstrates a good understanding of the subject matters and can demonstrate good performance in the individual assignments, group project and final examination with minor errors and omissions
B	The student demonstrates a reasonable understanding of the subject matters and can demonstrate reasonable level of performance in the individual assignments, group project and final examination
C	The student demonstrates a reasonable understanding of the subject matters and can demonstrate reasonable level of performance in the individual assignments, group project and final examination with some errors and omissions
D	The student fails to demonstrate a reasonable understanding of the subject matters and fails to demonstrate reasonable performance in the individual assignments, group project and final examination.
F	The student fails to demonstrate any understanding of the subject matters and shows poor performance in the individual assignments, group project and final examination.

7. Feedback for evaluation

CTE

8. Reading

Recommended Reading:

1. Storytelling with Data: A Data Visualization Guide for Business Professionals

2. The Visual Display of Quantitative Information
Edward Tufte, 1983

Knowledge Is Beautiful: Impossible Ideas, Invisible Patterns, Hidden Connections--
Visualized

David McCandless, 2014

Knowledge Is Beautiful: Impossible Ideas, Invisible Patterns, Hidden Connections--
Visualized

9. Course components

Activity	Hours/week
Lecture/Tutorial	3
Programming/Homework	3
Reading and Group Discussion	1

10. Indicative teaching plan

Week	Content/ topic/ activity
FINAL EXAM	Final Exam
LECTURE 1	Course Introduction: Data Visualization
LECT	Designing Dashboards and Interactive Visualizations

URE 10	
LECT URE 11	Automating Data Visualization
LECT URE 12	Project Presentation (I)
LECT URE 13	Project Presentation (II)
LECT URE 2	Logic, Beauty, and Principle in Data Visualization
LECT URE 3	Structuring Visual Narratives: Storytelling with Data
LECT URE 4	Data Preparation and Cleaning for Visualization
LECT URE 5	Basic Data Visualization and Tool Usage
LECT URE 6	Basic Data Visualization and Tool Usage
LECT URE 7	Mid-term Exam
LECT URE 8	Advanced Data Visualization and Tool Usage
LECT URE 9	Advanced Data Visualization and Tool Usage

11. **Implementation plan**

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections.

[Example: 150 students for lecture (x 2); 30 students for tutorials (x 10)]

The implementation plan may vary from year to year. Please indicate expected

enrolment, and number of sections.

80 students for lecture/tutorials

3 sessions in total

12. Any other information

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences?

Have the details (as in this document) been approved at School or other level in CUHK-Shenzhen?

Submit application

14. Version date

Version number	1
As of (date)	2025-07-31